



An adjoint-based method for liner impedance eduction: Validation and numerical investigation

Julien Primus*, Estelle Piot, Frank Simon

Onera—The French Aerospace Lab, F-31055 Toulouse, France

ARTICLE INFO

Article history:

Received 2 March 2012

Received in revised form

26 June 2012

Accepted 16 July 2012

Handling Editor: Y. Auregan

Available online 5 September 2012

ABSTRACT

An inverse method for liner acoustic impedance eduction in a duct is presented. The determination of adjoint equations allows to make explicit the directional derivatives of the objective function, needed by the search algorithm. Direct and adjoint system of equations are spatially discretized by a discontinuous Galerkin scheme. The propagation model and the eduction process are successfully validated on NASA benchmark data extracted from literature, in a no-flow configuration. Objective functions written in terms of acoustic pressure and acoustic velocity are considered successively. Special attention is given to the possibilities offered by the method, such as multiparameter identification, with the study of a non-homogeneous liner. Convergence history of the process is also examined, revealing interesting properties for the velocity-based objective function compared to the pressure-based objective function. In further developments, the method is planned to be used with acoustic velocity fields obtained by laser doppler velocimetry.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

Reduction of the noise generated by aircraft turbofans can be achieved by lining the nacelle with passive acoustic treatments, typically made of partitioned air cavities topped by a perforated sheet. Optimization of these liners efficiency (see for instance [1]) requires an accurate knowledge of their absorbing properties in the conditions of use, namely the presence of a grazing flow. These properties are represented by the acoustic impedance, an homogenized quantity defined locally as the ratio of the acoustic pressure at the surface over the normal acoustic velocity. This parameter, which is independent of the incidence angle for the so-called “locally reacting” liners, embraces the nonlinear phenomena occurring around the perforations, and due to the complex sound environment in the duct [2].

Alongside with the large variety of semi-empirical models available [3–5], various ways exist to measure the acoustic impedance in this context. The early work of Dean in 1974 [6] led to an *in situ* “two microphones” technique, widely used when it comes to characterizing single-layer liners. At Onera, liner impedance is deduced from velocity measurements acquired by laser doppler velocimetry (LDV) in an aero-thermo-acoustic bench [7]. Nevertheless the general trend is to favor indirect techniques, where the liner impedance is the parameter minimizing the difference between a given theoretical model and the measurements. The liner sample is then let undamaged by the introduction of any probe. Since the mid-1990s NASA has made considerable work in this field [8,9], developing and enhancing an impedance eduction method using finite element modeling and pressure measurements on the rigid walls or, lately, pressure gradient evaluation at the source and termination planes [10].

* Corresponding author.

E-mail addresses: julien.primus@onera.fr (J. Primus), estelle.piot@onera.fr (E. Piot), frank.simon@onera.fr (F. Simon).

Other examples of indirect approaches include the work of Aurégan et al. [11] or Elnady et al. [12] where the acoustic field in a three-zone duct (unlined-lined-unlined) is expanded in terms of the unlined duct eigenmodes and treated by a mode matching method. The minimization process relies on pressure measurements upstream and downstream of the lined section, the objective function being expressed either directly from pressure [12] or through the coefficients of the scattering matrix [11].

The mode matching methods cited above offer the advantage of taking high-order modes into account, but are derived from the Helmholtz equation and thus assume uniform flow. Richter [13] developed a time-domain eduction method based on the linearized Euler equations (LEE) and involving a five parameters extended Helmholtz resonator model. He showed the improvements brought by a better estimation of the measured flow profile. This prompted Eversman and Gallman [14] to include an effective mean flow Mach number in their eduction process. This one also involves the exit impedance as a search parameter, and the inverse method is performed with a finite element model. Inclusion of the effective Mach number and of the termination impedance contributes to a better data fit with measurements.

The eduction procedure displayed and benchmarked in the present paper is based on the 2D linearized Euler equations in the frequency domain, in order to account for sheared flow in further developments. These equations are solved on an unstructured mesh thanks to a discontinuous Galerkin formulation, which presents advantageous properties. First, it is weakly dispersive and dissipative, as demonstrated for example by Hu and Atkins for 2D propagating waves [15]. In addition, boundary conditions are imposed through fluxes, which is particularly robust and straightforward [16]. The computational cost, which is the main drawback of discontinuous Galerkin methods, is not of the utmost importance as we are concerned with small-sized meshes. The minimization of the objective function is achieved by the resolution, at each iteration on the liner impedance, of the direct and adjoint equations. Contrary to other methods such as finite differences, the use of the adjoint state leads to an analytic expression of the derivatives of interest, at a cost which remains independent of the number of search parameters [17].

Section 2 states the discontinuous Galerkin formulation of the direct problem, and gives a complete review of the boundary conditions used. It also shows how the use of a Lagrangian function allows to write the adjoint system, and the gradient needed for the identification process is formulated analytically from both direct and adjoint states. The consideration of other search parameters, such as source amplitude or exit reflection coefficient, is then made easy. In Section 3 the propagation model is first validated with experimental data from literature in a no-flow configuration, and the eduction process is benchmarked as well. Influence of the exit impedance and sensitivity of the objective function are analyzed. An objective function written in terms of acoustic pressure is used. Since the process will in further developments be based on acoustic velocity fields obtained from LDV at Onera, a velocity objective function is studied likewise. Finally, an illustration of the first-order condition for optimality with the adjoint state is given.

2. Direct and adjoint aeroacoustic formulations

2.1. Propagation model in lined ducts

2.1.1. Variational formulation

The computational model relies on the linearized Euler equations written under a non-conservative form, for an $e^{i\omega t}$ time dependence (ω is the angular frequency). In the no-flow configurations considered in this paper, the LEE could reduce to a convected wave equation, thus decreasing the number of variables. Nevertheless, the use of the LEE opens the possibility to consider a non-uniform flow profile in further studies, thus allowing the inclusion of refraction effects due to the shear layer.

Consider a subsonic steady mean flow propagating in a 2D duct (Fig. 1), with velocity U_0 along the streamwise coordinate x_1 and V_0 along the transversal coordinate x_2 , and submitted to an acoustic perturbation $\boldsymbol{\varphi} = (u, v, p)^\top$, where $^\top$ stands for the transpose. The vector $\mathbf{u} = (u, v)^\top$ represents the acoustic velocity, and p is the acoustic pressure. Let c_0 be the speed of sound and ρ_0 the density of the mean flow. Velocities are made dimensionless by c_0 , pressure by $\rho_0 c_0^2$, lengths by the duct height H and time by H/c_0 . In this paper, the density and speed of sound of the flow are assumed uniform. Although the numerical results presented in this paper deal with a no-flow configuration, governing equations are treated in a more general way and include the influence of an incompressible flow. The local form of the incompressibility equation is written

$$\partial_1 U_0 + \partial_2 V_0 = 0 \quad (1)$$

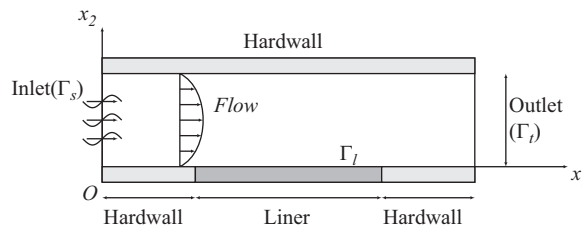


Fig. 1. Generic geometry of the studied system.

where ∂_i denotes the partial derivative with respect to the coordinate x_i .

The matrix system of governing equations then reads

$$j\omega\boldsymbol{\varphi} + \mathbf{A}^i \partial_i \boldsymbol{\varphi} + \mathbf{B}\boldsymbol{\varphi} = 0 \quad (2)$$

also noted $\mathbf{L}\boldsymbol{\varphi} = 0$. The repeated index indicates a sum over the possible values of i , following Einstein summation convention. Matrices $\mathbf{A}^i$ and $\mathbf{B}$ are defined by

$$\mathbf{A}^1 = \begin{pmatrix} U_0 & 0 & 1 \\ 0 & U_0 & 0 \\ 1 & 0 & U_0 \end{pmatrix}, \quad \mathbf{A}^2 = \begin{pmatrix} V_0 & 0 & 0 \\ 0 & V_0 & 1 \\ 0 & 1 & V_0 \end{pmatrix} \quad (3)$$

and

$$\mathbf{B} = \begin{pmatrix} \partial_1 U_0 & \partial_2 U_0 & 0 \\ \partial_1 V_0 & -\partial_1 U_0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (4)$$

Matrix $\mathbf{B}$ contains terms relative to the mean flow profile, and cancels out for a uniform mean flow. In these matrices the mean flow velocities U_0 and V_0 are dimensionless.

The physical domain Ω is now approximated by a computational domain Ω_h obtained by triangulation into non-overlapping elements Ω_e

$$\Omega \simeq \Omega_h = \bigcup_e \Omega_e \quad (5)$$

In order to state a discontinuous Galerkin (DG) formulation of the system, we consider a piecewise polynomial approximate solution $\boldsymbol{\varphi}_h$ of Eq. (2) on the computational domain Ω_h . First-order Lagrange elements are used. Due to the possible discontinuities across the elements boundaries, we are first interested in the restriction $\boldsymbol{\varphi}_h^e = \boldsymbol{\varphi}_h|_{\Omega_e}$ of $\boldsymbol{\varphi}_h$ to the element Ω_e (see Fig. 2). Multiplying Eq. (2) by a test function $\boldsymbol{\psi}_h^e$ and integrating over the element yields

$$\int_{\Omega_e} (j\omega\boldsymbol{\varphi}_h^e + \mathbf{A}^i \partial_i \boldsymbol{\varphi}_h^e + \mathbf{B}\boldsymbol{\varphi}_h^e) \cdot \boldsymbol{\psi}_h^e \, d\Omega_e = 0 \quad (6)$$

Let $\mathbf{n} = n_i x_i$ be the outward unit normal on the boundary $\partial\Omega_e$. A first integration by parts leads to

$$\int_{\Omega_e} (j\omega\boldsymbol{\varphi}_h^e + \mathbf{B}\boldsymbol{\varphi}_h^e) \cdot \boldsymbol{\psi}_h^e \, d\Omega_e - \int_{\partial\Omega_e} \mathbf{A}^i \partial_i \boldsymbol{\psi}_h^e \cdot \boldsymbol{\varphi}_h^e \, d\Omega_e + \sum_{k=1}^3 \int_{\partial\Omega_{e,k}} \mathbf{A}^i n_i \boldsymbol{\varphi}_h^e \cdot \boldsymbol{\psi}_h^e \, d\Gamma = 0 \quad (7)$$

Discontinuities being allowed at interfaces, the integrand on each edge $\partial\Omega_{e,k}$ ($k \in \llbracket 1, 3 \rrbracket$) of the triangle is multivalued and its meaning needs to be precised. To do that, let us define the interior and exterior traces $\boldsymbol{\varphi}_h^{e-}$ and $\boldsymbol{\varphi}_h^{e+}$ of $\boldsymbol{\varphi}_h^e$ at $\mathbf{x} \in \partial\Omega_e$

$$\begin{cases} \boldsymbol{\varphi}_h^{e-}(\mathbf{x}) = \lim_{\mathbf{x}' \rightarrow \mathbf{x}} \boldsymbol{\varphi}_h^e(\mathbf{x}') \\ \boldsymbol{\varphi}_h^{e+}(\mathbf{x}) = \lim_{\mathbf{x}' \rightarrow \mathbf{x}} \boldsymbol{\varphi}_h^e(\mathbf{x}') \end{cases} \quad (8)$$

We then introduce a numerical flux $\widehat{\mathbf{F}}_e(\boldsymbol{\varphi}_h^{e-}, \boldsymbol{\varphi}_h^{e+})$ which will ensure the connection between elements and is therefore a function of the interior and exterior traces. At this point, the variational formulation (7) can be written

$$\int_{\Omega_e} (j\omega\boldsymbol{\varphi}_h^e + \mathbf{B}\boldsymbol{\varphi}_h^e) \cdot \boldsymbol{\psi}_h^e \, d\Omega_e - \int_{\partial\Omega_e} \mathbf{A}^i \partial_i \boldsymbol{\psi}_h^e \cdot \boldsymbol{\varphi}_h^e \, d\Omega_e + \sum_{k=1}^3 \int_{\partial\Omega_{e,k}} \widehat{\mathbf{F}}_e(\boldsymbol{\varphi}_h^{e-}, \boldsymbol{\varphi}_h^{e+}) \cdot \boldsymbol{\psi}_h^{e-} \, d\Gamma = 0 \quad (9)$$

The numerical flux will be built explicitly further, but we can already note that it must satisfy the consistency and conservativity properties, that is

$$\widehat{\mathbf{F}}_e(\boldsymbol{\varphi}, \boldsymbol{\varphi}) = \mathbf{A}^i n_i \boldsymbol{\varphi} \quad \text{and} \quad \widehat{\mathbf{F}}_e(\boldsymbol{\varphi}_h^{e-}, \boldsymbol{\varphi}_h^{e+}) + \widehat{\mathbf{F}}_{e'}(\boldsymbol{\varphi}_h^{e'-}, \boldsymbol{\varphi}_h^{e'+}) = 0 \quad (10)$$

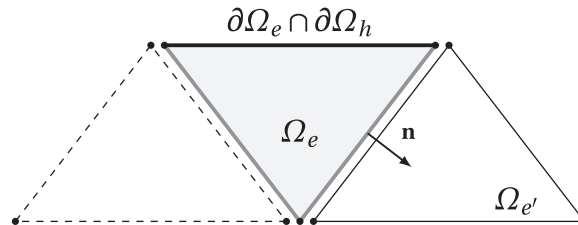


Fig. 2. An element Ω_e adjacent to the computation domain boundary.

From there we can integrate (9) by parts once again, but now that the link between elements is ensured we consider that $\varphi_h^e|_{\partial\Omega_{e,k}} = \varphi_h^{e-}$. This yields

$$\int_{\Omega_e} (j\omega\varphi_h^e + \mathbf{A}^i \partial_i \varphi_h^e + \mathbf{B}\varphi_h^e) \cdot \psi_h^e d\Omega_e + \sum_{k=1}^3 \int_{\partial\Omega_{e,k}} [\widehat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) - \mathbf{A}^i n_i \varphi_h^{e-}] \cdot \psi_h^{e-} d\Gamma = 0 \quad (11)$$

Of course, comparison of Eqs. (6) and (11) indicates that the summation term in (11) is equal to zero for the solution, which is correlated to the consistency property of the numerical flux stated in Eq. (10).

This formulation has to be completed with an explicit expression of the numerical flux and with the specification of imposed boundary conditions. To obtain the final variational formulation, Eq. (11) is summed over all the elements Ω_e .

2.1.2. Numerical flux and boundary conditions

As it links neighboring elements together and allows the application of the boundary conditions, the choice of numerical flux is a crucial step in the construction of the DG scheme. Different cases arise for the construction of $\widehat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+})$ and the calculation of the corresponding integral in Eq. (11), depending on whether the edge $\partial\Omega_{e,k}$ is in contact with the boundary or not, and on the kind of boundary condition possibly involved. The types of numerical flux chosen hereafter guarantee that the scheme built in this way for the unsteady problem obtained by inverse Fourier transform of Eq. (2) is stable and convergent [18]. Even though the well-posedness of the scheme developed here for the harmonic problem (2) could not be proven, it has been shown in previous works that for no-flow, uniform flow and sheared flow configurations no particular difficulty should be encountered [19].

- If $\partial\Omega_{e,k}$ is an internal edge, an upwind numerical flux is used. To establish a one-sided scheme stable for the unsteady problem, several upstream-differencing techniques are available. One can refer to the review of Harten et al. [20] where the authors pay special attention to the Godunov-type schemes, but also to the flux-vector splitting method (FVS) introduced by Steger and Warming [21] for the conservation-law form of the gas dynamics equations, and later applied for example by Gabard [22]. In the case of Eq. (2), where the Euler equations are linearized and in non-conservative form, these techniques are equivalent [23]. The key idea of flux-splitting is to use the hyperbolicity of the system, which allows to decompose the solution on the boundary $\partial\Omega_{e,k}$ into incoming and outgoing waves, the so-called characteristics. Mathematically, this is linked to the fact that the matrix $\mathbf{A}^i n_i$ is real symmetric and can be split into matrices $\mathbf{A}^i n_i^+$ and $\mathbf{A}^i n_i^-$, respectively, associated to the positive and negative eigenvalues such as

$$\mathbf{A}^i n_i = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1} = \mathbf{P}\mathbf{\Lambda}^+ \mathbf{P}^{-1} + \mathbf{P}\mathbf{\Lambda}^- \mathbf{P}^{-1} = \mathbf{A}^i n_i^+ + \mathbf{A}^i n_i^- \quad (12)$$

where $\mathbf{P}$ is the matrix of the eigenvectors of $\mathbf{A}^i n_i$, and $\mathbf{\Lambda} = \mathbf{\Lambda}^+ + \mathbf{\Lambda}^-$ the diagonal matrix formed by the eigenvalues of $\mathbf{A}^i n_i$, which are the phase speed of the characteristics. Matrices $\mathbf{\Lambda}^+$ and $\mathbf{\Lambda}^-$ are diagonal and contain the positive and negative eigenvalues, respectively. These eigenvalues being $U^i n_i, (U^i n_i + 1)$ and $(U^i n_i - 1)$, the expression of $\mathbf{A}^i n_i^+$ and $\mathbf{A}^i n_i^-$ will be function of the sign of the local normal flow velocity $U_n = U^i n_i$. Calculations give the following general formula for $\mathbf{A}^i n_i^-$:

$$\mathbf{A}^i n_i^- = \inf(0, U_n) \begin{pmatrix} n_2^2 & -n_1 n_2 & 0 \\ -n_1 n_2 & n_1^2 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{U_n - 1}{2} \begin{pmatrix} n_1^2 & n_1 n_2 & -n_1 \\ n_1 n_2 & n_2^2 & -n_2 \\ -n_1 & -n_2 & 1 \end{pmatrix} \quad (13)$$

From there, an upwind numerical flux is achieved by associating the interior trace to outgoing waves (i.e. positive eigenvalues), and the exterior trace to ingoing waves (negative eigenvalues), as indicated by Fig. 3. Formally,

$$\widehat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) = \mathbf{A}^i n_i^+ \varphi_h^{e-} + \mathbf{A}^i n_i^- \varphi_h^{e+} \quad (14)$$

Such a scheme ensures low dispersion errors [19]. The corresponding integral in Eq. (11) for an internal edge $\partial\Omega_{e,k}$ then reads, given the splitting (12):

$$\int_{\partial\Omega_{e,k} \cap \Gamma_i} [\widehat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) - \mathbf{A}^i n_i \varphi_h^{e-}] \cdot \psi_h^{e-} d\Gamma = \int_{\partial\Omega_{e,k} \cap \Gamma_i} \mathbf{A}^i n_i^- (\varphi_h^{e+} - \varphi_h^{e-}) \cdot \psi_h^{e-} d\Gamma \quad (15)$$

where Γ_i is the set of internal edges of the element Ω_e . This expression combined with the conservativity property (10) can be seen as a Rankine–Hugoniot condition across a fixed discontinuity.

- If $\partial\Omega_{e,k}$ is a boundary edge, three different boundary conditions are to be taken into account. In the generic flow duct configuration considered (Fig. 1), a source condition (typically a plane wave) is applied at the inlet on the ID boundary Γ_s . The presence of the liner and of rigid walls (equivalent to a liner with infinite impedance) is modeled by the

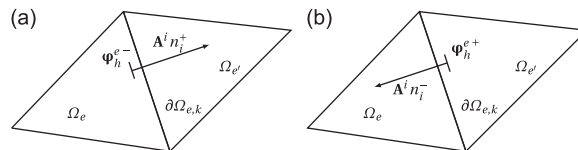


Fig. 3. Upwind numerical flux: (a) outgoing and (b) ingoing characteristics of the element Ω_e .

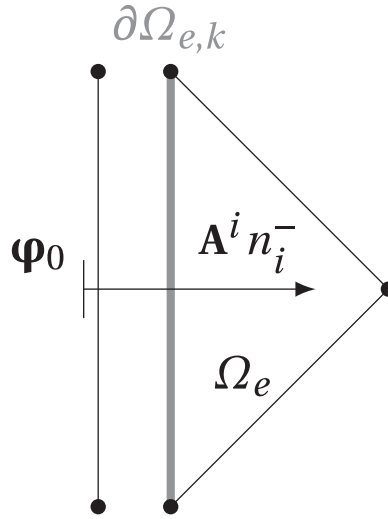


Fig. 4. Source boundary condition.

implementation of an impedance boundary condition. The outlet Γ_t can either be submitted to a non-reflective boundary condition or an exit impedance boundary condition.

○ *Source boundary condition*

A source boundary condition expressed physically as $\varphi = \varphi_0$ is imposed through the introduction of 1D fictitious cells, for which the state vector is φ_0 and constitutes the exterior trace for the element Ω_e (see Fig. 4). The edge between the fictitious cell and the current element is then considered as an internal edge, which yields

$$\int_{\partial\Omega_{e,k} \cap \Gamma_s} [\hat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) - \mathbf{A}^i n_i \varphi_h^{e-}] \cdot \psi_h^{e-} d\Gamma = \int_{\partial\Omega_{e,k} \cap \Gamma_s} \mathbf{A}^i n_i^- (\varphi_0 - \varphi_h^{e-}) \cdot \psi_h^{e-} d\Gamma \quad (16)$$

○ *Non-reflecting boundary condition*

To avoid reflections at open boundaries of the computational domain, characteristic boundary conditions derived from the upwind flux splitting (12) are used [24]. The incoming waves related to matrix $\mathbf{A}^i n_i^-$ are suppressed, leading to

$$\hat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) = \mathbf{A}^i n_i^+ \varphi_h^{e-} \quad (17)$$

and the boundary integral term for transparent conditions is

$$\int_{\partial\Omega_{e,k} \cap \Gamma_t} [\hat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) - \mathbf{A}^i n_i \varphi_h^{e-}] \cdot \psi_h^{e-} d\Gamma = \int_{\partial\Omega_{e,k} \cap \Gamma_t} -\mathbf{A}^i n_i^- \varphi_h^{e-} \cdot \psi_h^{e-} d\Gamma \quad (18)$$

○ *Impedance boundary condition*

Similar works [9,11,12] often assume a uniform flow profile, the inverse method being built on the Helmholtz equation. Therefore such approaches require a lined wall boundary condition established from the continuity of transverse acoustical displacement at the surface, the so-called Ingard–Myers boundary condition [25]. More recent works investigate the effect of shear flow, still with this boundary condition. Moreover Renou and Aurégan [26] lately highlighted limits of this formulation and proposed a modified Ingard–Myers boundary condition which holds whether the acoustic propagation in the flow duct is downstream or upstream. As regards our process, the choice of a standard impedance condition has been made since final applications will aim at acquiring the sheared flow properties with laser Doppler velocimetry measurements. The implementation of the impedance boundary condition thus relies on the classical normalized specific impedance z , that is

$$z = \frac{p}{v^i n_i} = \frac{p}{v_n} \quad (19)$$

with $v_n = v^i n_i$ the normal acoustic velocity.

Here again a fictitious 1D cell is used to represent the exterior trace for Ω_e . The chosen numerical scheme is centered and involves the average of φ on the boundary, defined as $\llbracket \varphi \rrbracket = (\varphi_h^{e-} + \varphi_h^{e+})/2$. The numerical flux is then $\hat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) = \mathbf{A}^i n_i \llbracket \varphi \rrbracket$ and

$$\int_{\partial\Omega_{e,k} \cap \Gamma_t} [\hat{\mathbf{F}}_e(\varphi_h^{e-}, \varphi_h^{e+}) - \mathbf{A}^i n_i \varphi_h^{e-}] \cdot \psi_h^{e-} d\Gamma = \int_{\partial\Omega_{e,k} \cap \Gamma_t} \mathbf{A}^i n_i \frac{\varphi_h^{e+} - \varphi_h^{e-}}{2} \cdot \psi_h^{e-} d\Gamma \quad (20)$$

The exterior trace is imposed in such a way that the impedance boundary condition (19) is accounted for with respect to the interior trace

$$\begin{cases} v_n^+ = \frac{1}{z} p^- \\ p^+ = z v_n^- \end{cases} \quad (21)$$

where the $+$ and $-$ superscripts still stand for the exterior and interior trace of the quantity. Quick calculations give

$$\int_{\partial\Omega_{e,k} \cap \Gamma_l} [\widehat{\mathbf{F}}_e(\boldsymbol{\varphi}_h^{e-}, \boldsymbol{\varphi}_h^{e+}) - \mathbf{A}^i n_i \boldsymbol{\varphi}_h^{e-}] \cdot \boldsymbol{\psi}_h^{e-} d\Gamma = \int_{\partial\Omega_{e,k} \cap \Gamma_l} \mathbf{M}_z \boldsymbol{\varphi}_h^{e-} \cdot \boldsymbol{\psi}_h^{e-} d\Gamma \quad (22)$$

The matrix $\mathbf{M}_z$ is defined as

$$\mathbf{M}_z = \frac{1}{2} \begin{pmatrix} zn_1^2 & zn_1 n_2 & -n_1 \\ zn_1 n_2 & zn_2^2 & -n_2 \\ -n_1 & -n_2 & 1/z \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \mathbf{z}\mathbf{n} \otimes \mathbf{n} & -\mathbf{n} \\ -\mathbf{n}^\top & 1/z \end{pmatrix} \quad (23)$$

Previous studies [19] attested that a formulation bringing into play the reflection coefficient $\beta = (z-1)/(z+1)$ is equivalent but numerically preferable, resulting in

$$\int_{\partial\Omega_{e,k} \cap \Gamma_l} [\widehat{\mathbf{F}}_e(\boldsymbol{\varphi}_h^{e-}, \boldsymbol{\varphi}_h^{e+}) - \mathbf{A}^i n_i \boldsymbol{\varphi}_h^{e-}] \cdot \boldsymbol{\psi}_h^{e-} d\Gamma = \int_{\partial\Omega_{e,k} \cap \Gamma_l} \mathbf{M}_\beta \boldsymbol{\varphi}_h^{e-} \cdot \boldsymbol{\psi}_h^{e-} d\Gamma \quad (24)$$

with

$$\mathbf{M}_\beta = \frac{1}{2} \begin{pmatrix} (\beta+1)\mathbf{n} \otimes \mathbf{n} & -(1-\beta)\mathbf{n} \\ -(1+\beta)\mathbf{n}^\top & (1-\beta) \end{pmatrix} \quad (25)$$

Writing $\mathbf{M}_\beta$ in this way allows to write the strong form of the boundary condition $\mathbf{M}_\beta \boldsymbol{\varphi} = 0$ either for a liner or for a rigid wall (by taking $\beta = 1$) without singularity issues.

The boundary conditions can also be written under a strong form. The complete system then reads

$$\begin{cases} \mathbf{L}\boldsymbol{\varphi} = 0 & \text{on } \Omega \\ \mathbf{M}_\beta \boldsymbol{\varphi} = 0 & \text{on } \Gamma_l \\ \mathbf{A}^i n_i^- \boldsymbol{\varphi} = 0 & \text{on } \Gamma_t \\ \mathbf{A}^i n_i^- \boldsymbol{\varphi} = \mathbf{A}^i n_i^- \boldsymbol{\varphi}_0 & \text{on } \Gamma_s \end{cases} \quad (26)$$

2.2. Adjoint problem and objective function derivation

The impedance eduction process consists in solving an inverse problem of parameters identification, by minimizing an objective function $\mathcal{J}$ representative of the distance between the DG numerical simulation and the measurement data. This objective function can be written under a general form

$$\mathcal{J} = \int_{\Omega} \|\boldsymbol{\varphi}_{\text{calc}} - \boldsymbol{\varphi}_{\text{meas}}\|^2 I_{\Omega_{\text{obs}}}(\mathbf{x}) \quad (27)$$

where $I_{\Omega_{\text{obs}}}(\mathbf{x})$ is the indicator function of Ω_{obs} , equal to unity if $\mathbf{x} \in \Omega_{\text{obs}}$ and to zero otherwise. Depending on the case dealt with, only some components of $\boldsymbol{\varphi}$ will be included. The observation region Ω_{obs} can be discrete (for example a given set of microphone positions) or continuous.

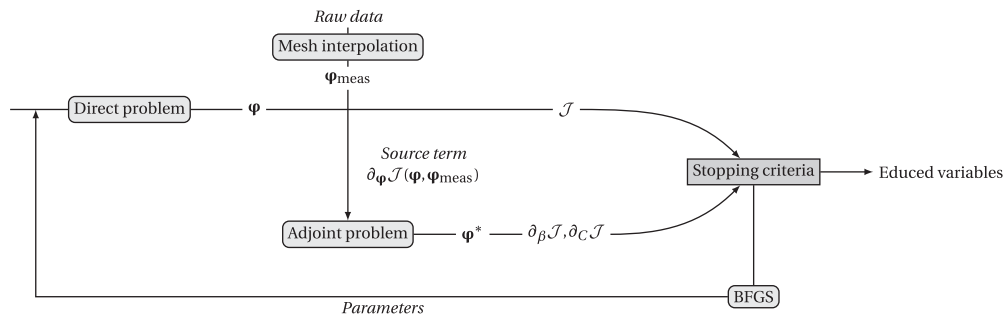


Fig. 5. Summary scheme of the eduction loop.

During the iterative search process (Fig. 5) the value, gradient and approximated Hessian matrix of the objective function at the current step are used to determine a new set of parameters which should lead to a smaller objective function, that is a better fit of the numerical results to the experimental data.

Even though we are primarily interested in the liner impedance, other parameters can be included in the search process to improve the reliability of the process. First, even when the duct is equipped with a quasi-anechoic termination, reflecting exit impedances can be measured [27]. This suggests to allow the introduction of an exit impedance as a search parameter instead of the non-reflecting boundary condition presented in Section 2.1.2. Furthermore, the source term having an arbitrary unit amplitude in the DG code, a complex calibration coefficient is also considered.

The gradient of the objective function can be calculated by various means amongst which the finite difference method seems the most natural approach. However, it results in an approximate assessment of the gradient, and the cost of the evaluation increases with the number of search parameters. We opt for an adjoint-based calculation of the gradient because it leads to an analytical expression of the directional derivatives, and allows to test non-homogeneous liners represented by several impedances without additional computational cost. Indeed, whatever the number of search parameters, the computation of directional derivatives only requires the computation of the direct and adjoint solutions. This property makes the use of adjoint equations popular in the field of shape optimization [28]. The use of the adjoint equations is also widely found in the study of diverse problems such as sensitivity [28,29], boundary layer receptivity [29,30], or jet noise control [31,32].

Two options turn up to obtain a discretized adjoint formulation: one can choose the “differentiate-then-discretize” possibility or the “discretize-then-differentiate” one [33]. In the first one, the adjoint equations and the objective function derivatives with respect to the parameters are obtained from the direct continuous problem, and the adjoint equations are discretized subsequently with the DG scheme. In the second one on the contrary, the first step is to discretize the continuous direct equations and to differentiate afterwards to get the discretized adjoint system and the objective function derivatives. Since the “differentiate” and “discretize” operations do not commute in most cases, these approaches are usually not equivalent. However, it has been proven [34] that with the DG formulation presented previously, they do commute and the two techniques can be used equally. The choice has thus been made to use the “differentiate-then-discretize” approach.

To find the adjoint system of equations and the directional derivatives of $\mathcal{J}$, the method of Lagrange multipliers is used. A Lagrangian function is thus built from the original objective function, considering the direct equations and the associated boundary conditions as constraints. The objective function $\mathcal{J}$ is usually expressed as a function of the state vector $\boldsymbol{\varphi}$, and is indirectly dependent of the search parameters, namely the impedance (denoted z in the general case) and the calibration coefficient C ($C \in \mathbb{C}$) mentioned above. This dependence is noted $\mathcal{J}(\boldsymbol{\varphi}, z, C)$. The Lagrangian will be a function of the same variables, but also of the Lagrange multiplier $\boldsymbol{\varphi}^*$ which will further be defined as the solution of the adjoint equations.

Let $\langle \cdot, \cdot \rangle$ be the complex inner product. The Lagrangian function is expressed in the following way (the e superscript and h subscript are dropped for clarity of notation):

$$\begin{aligned} \mathcal{L}(\boldsymbol{\varphi}, z, C, \boldsymbol{\varphi}^*) = & \mathcal{J}(\boldsymbol{\varphi}, z, C) - [\langle \mathbf{L}\boldsymbol{\varphi}, \boldsymbol{\varphi}^* \rangle_{\Omega} + \langle \mathbf{A}^i n_i^- (\boldsymbol{\varphi}^+ - \boldsymbol{\varphi}^-), \boldsymbol{\varphi}^* \rangle_{\Gamma_i} \\ & + \langle \mathbf{A}^i n_i^- (C\boldsymbol{\varphi}_0 - \boldsymbol{\varphi}^-), \boldsymbol{\varphi}^* \rangle_{\Gamma_s} + \langle \mathbf{M}_{\beta} \boldsymbol{\varphi}^-, \boldsymbol{\varphi}^* \rangle_{\Gamma_l} + \langle -\mathbf{A}^i n_i^- \boldsymbol{\varphi}^-, \boldsymbol{\varphi}^* \rangle_{\Gamma_t}] \end{aligned} \quad (28)$$

Note that the complex conjugate should be added to ensure that the Lagrangian is real-valued, but has been omitted for the sake of clarity. The Lagrangian is constructed in such a way that it is equal to the objective function when the direct equations are fulfilled. A small variation of $\mathcal{L}(\boldsymbol{\varphi}, z, C, \boldsymbol{\varphi}^*)$ can be written

$$\begin{aligned} \delta \mathcal{L}(\boldsymbol{\varphi}, z, C, \boldsymbol{\varphi}^*) = & \delta \mathcal{J}(\boldsymbol{\varphi}, z, C) - [\langle \delta(\mathbf{L}\boldsymbol{\varphi}), \boldsymbol{\varphi}^* \rangle_{\Omega} + \langle \delta(\mathbf{A}^i n_i^- (\boldsymbol{\varphi}^+ - \boldsymbol{\varphi}^-)), \boldsymbol{\varphi}^* \rangle_{\Gamma_i} \\ & + \langle \delta(\mathbf{A}^i n_i^- (C\boldsymbol{\varphi}_0 - \boldsymbol{\varphi}^-)), \boldsymbol{\varphi}^* \rangle_{\Gamma_s} + \langle \delta(\mathbf{M}_{\beta} \boldsymbol{\varphi}^-), \boldsymbol{\varphi}^* \rangle_{\Gamma_l} + \langle \delta(-\mathbf{A}^i n_i^- \boldsymbol{\varphi}^-), \boldsymbol{\varphi}^* \rangle_{\Gamma_t}] \end{aligned} \quad (29)$$

The term related to the variation $\delta \boldsymbol{\varphi}^*$ of the Lagrange multiplier has been omitted, as it cancels out when the direct system of equations is satisfied. From there, solving the initial inverse problem on $\mathcal{J}$ comes down to searching a critical point for $\mathcal{L}$. The key idea is to factorize Eq. (29) by means of integrations by parts, so as to write this variation under the form

$$\delta \mathcal{L}(\boldsymbol{\varphi}, z, C, \boldsymbol{\varphi}^*) = \langle \partial_{\boldsymbol{\varphi}} \mathcal{L}, \delta \boldsymbol{\varphi} \rangle + \langle \partial_{\boldsymbol{\varphi}^*} \mathcal{L}, \delta \boldsymbol{\varphi}^* \rangle + \langle \partial_z \mathcal{L}, \delta z \rangle + \langle \partial_C \mathcal{L}, \delta C \rangle \quad (30)$$

and to cancel each directional derivative, assuming that the variables vary independently.

It has already been mentioned that the derivative with respect to the Lagrange multiplier is zero when the direct LEE and the associated boundary conditions are respected. The cancellation of the $\boldsymbol{\varphi}$ -derivative $\partial_{\boldsymbol{\varphi}} \mathcal{L}$ helps to state the system of adjoint equations, which in turn allows to express $\partial_z \mathcal{L}$ and $\partial_C \mathcal{L}$.

2.2.1. Derivation of the adjoint system

By integrating Eq. (28) by parts, it is found that the derivative with respect to the state vector is zero when the Lagrange multiplier $\boldsymbol{\varphi}^*$ satisfies the system

$$\begin{cases} \mathbf{L}^* \boldsymbol{\varphi}^* = \partial_{\boldsymbol{\varphi}} \mathcal{J} & \text{on } \Omega \\ \mathbf{M}_{\beta}^* \boldsymbol{\varphi}^* = 0 & \text{on } \Gamma_l \\ \mathbf{A}^i n_i^+ \boldsymbol{\varphi}^* = 0 & \text{on } \Gamma_t \cup \Gamma_s \end{cases} \quad (31)$$

where the operator $\mathbf{L}^* \cdot = -j\omega \cdot -\partial_i(\mathbf{A}^{i\dagger} \cdot) + \mathbf{B}^\dagger \cdot$ is the adjoint operator of the linearized Euler equations ($\dagger$ denotes the Hermitian conjugate). The Lagrange multiplier $\boldsymbol{\varphi}^*$ is called the *adjoint vector* or *adjoint state*. Matrix $\mathbf{M}_\beta^* = \mathbf{A}^i n_i + \mathbf{M}_\beta^\dagger$ is the adjoint impedance matrix, translating the adjoint impedance condition $p^* = -\bar{z} v^{*i} n_i$ with $\bar{z}$ the complex conjugate of z . It is worth noting that the minus sign in the multiplicative term $-j\omega$ accounts to an integration run backward in time in the temporal domain. As the source term $\partial_\varphi \mathcal{J}$ involves $\boldsymbol{\varphi}$, the direct system has to be solved before considering the adjoint system. The support of $\mathcal{J}$ being Ω_{obs} , this region will constitute the source localization for the adjoint problem.

2.2.2. Sensitivity to the search parameters

The stationarity of $\mathcal{L}$ with respect to the search parameters leads to the following expressions for the derivatives of the objective function:

$$\frac{\partial \mathcal{J}(\boldsymbol{\varphi}, z, C)}{\partial z_l} = - \left\langle \frac{\partial \mathbf{M}_\beta}{\partial \beta} \frac{\partial \beta}{\partial z_l} \boldsymbol{\varphi}, \boldsymbol{\varphi}^* \right\rangle_{\Gamma_l} \quad (32)$$

and

$$\frac{\partial \mathcal{J}(\boldsymbol{\varphi}, z, C)}{\partial C} = \langle \mathbf{A}^i n_i^- \boldsymbol{\varphi}_0, \boldsymbol{\varphi}^* \rangle_{\Gamma_s} \quad (33)$$

Vector $\boldsymbol{\varphi}_0$ is the source vector for the direct problem defined in Section 2.1.2.

As Eq. (32) remains general, this formula can be applied to any impedance and is not restricted to the liner impedance z_l . This namely allows to reduce the exit impedance z_t (leading to a scalar product over Γ_t instead of Γ_l), as well as to split the liner into several parts with different impedances to check the homogeneity property of the material.

2.2.3. Education loop

Finally, each iteration of the search process is made of the following steps:

1. The direct problem is solved with input z and C initial values.
2. The raw data from measurements are treated to get $\boldsymbol{\varphi}_{\text{meas}}$ on the computational domain. $\mathcal{J}$ is computed thanks to Eq. (27).
3. Combined to the state vector $\boldsymbol{\varphi}$ obtained at the first step, $\boldsymbol{\varphi}_{\text{meas}}$ enables the construction of the adjoint source term $\partial_\varphi \mathcal{J}$ by differentiating Eq. (27).
4. The adjoint problem is solved, for a computational cost equivalent to the direct problem.
5. Formula (32) and (33) are applied to calculate the gradient of the objective function.
6. The value of $\mathcal{J}$ and its gradient are submitted to the termination criteria. The first one stops the run if a change of the objective function between two loops k and $k+1$ is sufficiently small, that is [35]

$$\frac{\mathcal{J}_k - \mathcal{J}_{k+1}}{\max(|\mathcal{J}_{k+1}|, |\mathcal{J}_k|, 1)} \leq T_{\mathcal{J}} * \epsilon \quad (34)$$

where $T_{\mathcal{J}}$ is a parameter controlled by the user and ϵ is the machine precision, generated by the routine. The second stopping criterion terminates the run when the gradient projection has a sufficiently small infinity norm

$$\|\text{proj}(\nabla \mathcal{J})\|_\infty \leq T_\nabla \quad (35)$$

with T_∇ a parameter controlled by the user. The choice of $T_{\mathcal{J}}$ and T_∇ allows to adjust the precision of the final result. If one of the criteria (34) or (35) is satisfied, the current set of parameters is retained as the educed one. Otherwise $\mathcal{J}$ and the gradient are used as input of a BFGS-B algorithm (Broyden, Fletcher, Goldfarb, Shanno for bounded variables [36]) to get a new set of parameters z and C and loop to step 1.

3. Benchmark of the impedance education process

3.1. Validation of the direct propagation model

Before validating the education process, the reliability of the direct model needs to be confirmed. Although not presented here, results on academic test cases such as rigid walls or impedance tube configurations perfectly match the corresponding analytic solution. Besides, a previous paper [37] already compared favorably to results published by Özyörük et al. [38].

Here is presented the comparison of results obtained with our propagation code to benchmark data given by Jones et al. [27], in the no-flow configuration which will also be used in Section 3.2 to assess the robustness of the impedance education process. A scheme of the test section of the NASA grazing incidence tube (GIT) is given in Fig. 6. In the following sections, results will be presented dimensionalized.

Sound pressure levels (SPLs) and phases at the upper rigid wall are measured in [27] by 31 microphone probes. The liner and exit impedance used as input of our DG code are the one educed in the reference paper. The triangular mesh consists of Lagrange $\mathbb{P}_1$ elements. The frequencies considered are taken in the 0.5 to 3 kHz frequency range, with a

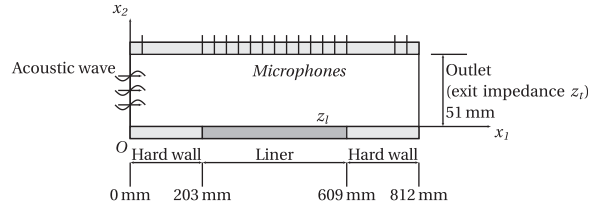


Fig. 6. 2D scheme of the test section of the NASA grazing incidence tube.

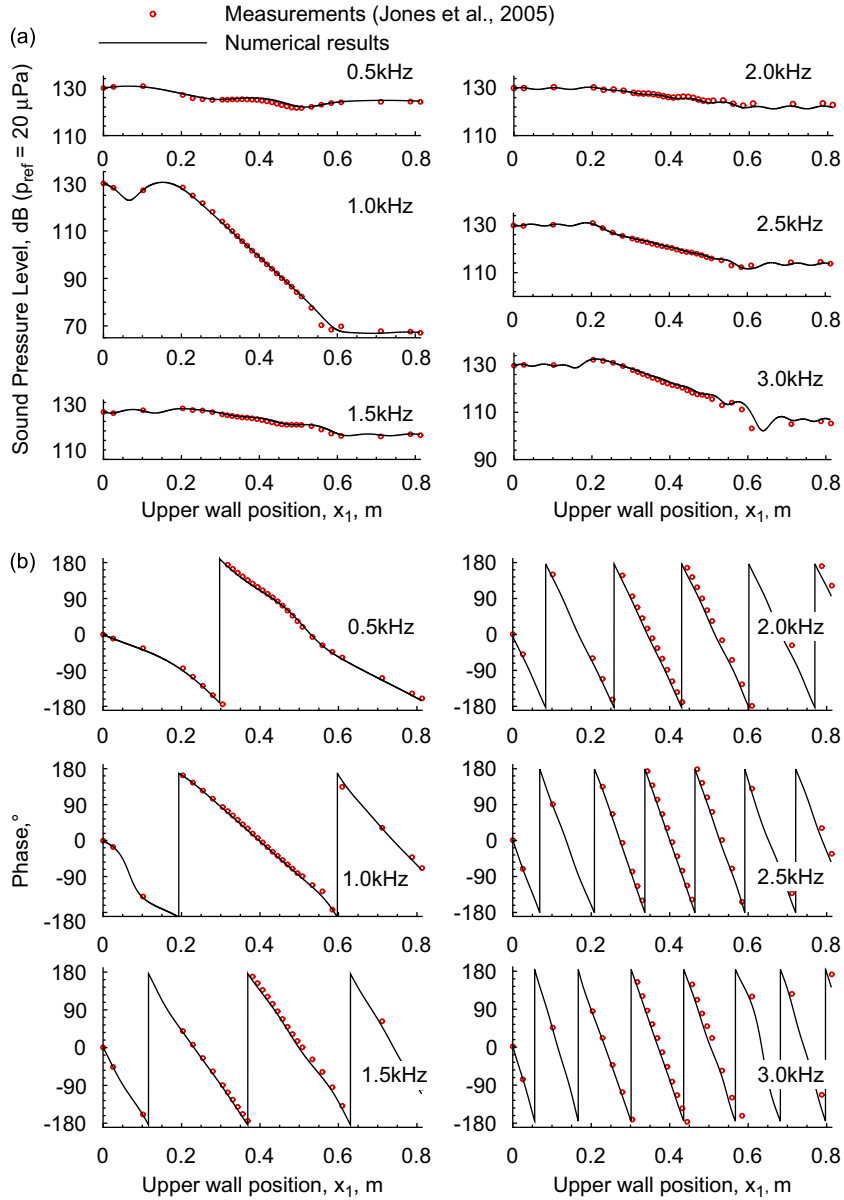


Fig. 7. (a) SPL and (b) phase on the upper wall of the NASA GIT. Measured values from Ref. [27] and simulation with the DG code.

500 Hz step. Mesh resolution has been chosen to reach 20 elements per wavelength at 3 kHz, which has been found sufficient to avoid numerical dissipation. Since arbitrary amplitude and phase are taken at the source plane in the DG code, simulated values are calibrated to match the measured ones at $x_1 = 0$ mm. As can be seen in Fig. 7, measured SPLs and phases are well captured by the DG propagation model.

3.2. Validation of the adjoint-based eduction method

3.2.1. Impedance eduction from wall pressure measurement

Benchmarking of the eduction process has been done on the data described above, taken from Ref. [27], where the liner impedance has been educed from a finite element model of the convected Helmholtz equation, the exit impedance being measured by a two-microphone method. To make a pertinent comparison, the same kind of objective function has been taken as the one used in [27], expressed as the squared error between the measured acoustic pressure p_{meas}^m and the simulated acoustic pressure p_{calc}^m , summed over the $N=31$ microphones

$$\mathcal{J}(\boldsymbol{\varphi}, \mathbf{z}, C) = \sum_{m=1}^N \|p_{\text{calc}}^m - p_{\text{meas}}^m\|^2 \quad (36)$$

The eduction process has been performed in two distinct configurations.

- The “NRp case” (for “non-reflective case” from pressure measurements) excludes the exit impedance of the search process. The non-reflective boundary condition (18) is applied at the termination plane, which boils down to a termination impedance $z_t = 1.0 + 0.0j$.
- The “Elp case” (for “exit impedance case” from pressure measurements) includes the exit impedance at the termination plane as a search parameter. The educed exit impedance is expected to be close to $z_t = 1.0 + 0.0j$, and this value is consequently taken as initial value for the search process. Results for the educed exit impedance are shown in Fig. 8, and agree very well with NASA measurements.

Searched impedances have been bounded such as

$$\begin{cases} 0.0 \leq r \leq 5.0 & \text{and } -5.0 \leq \chi \leq 5.0 & \text{for the liner} \\ 0.5 \leq r \leq 1.5 & \text{and } -0.5 \leq \chi \leq 0.5 & \text{for the exit impedance (“Elp case”)} \end{cases} \quad (37)$$

The calibration coefficient C is not bounded. The initial values are $z_{\text{init}} = 1.0 + 1.0j$ for the liner impedances and $C_{\text{init}} = 10.0 + 0.0j$ for the calibration coefficient.

Educed liner impedances are presented in Fig. 9 and provided in Appendix A (Tables A1 and A2). Taking the exit impedance into account (“Elp” case) gives a better agreement with the impedances from NASA, as already noticed by Eversman and Gallman [14]. There is hardly any difference between the impedances educed with our DG code and the reference impedances by Jones et al., except maybe at 2 kHz where a disagreement appears. A look at the reconstructed pressure at the upper wall (Fig. 10) reveals that there is only little difference between fields calculated by the DG code with the liner impedance value educed in the “Elp” or “NRp” case, and with the impedance educed by NASA. Jones et al. [27] have already noticed this kind of behavior near the anti-resonance frequency, namely between their classic 2D-FEM code and a quasi-3D FEM approach. A detailed investigation [39] revealed that these anomalies are not due to the grid refinement or to the choice of optimization algorithm, and that they are present whether the Helmholtz equation or linearized Euler equations are used. Recent work at the NASA Langley Research Center intends to increase confidence in the educed acoustic resistance, by making the anti-resonance frequency vary thanks to different cavity depths [40]. However, at the anti-resonance frequency the liner has very poor attenuation properties and tends to behave as a rigid wall. Consequently the differences in the educed impedances are of little importance, the most interesting behavior occurring around the resonance frequency.

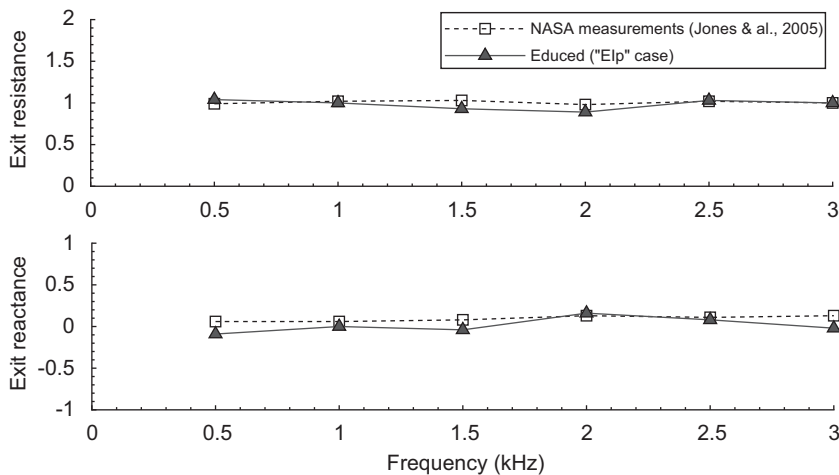


Fig. 8. Exit impedance educed from upper wall pressure measurements on the NASA GIT for the “Elp case”.

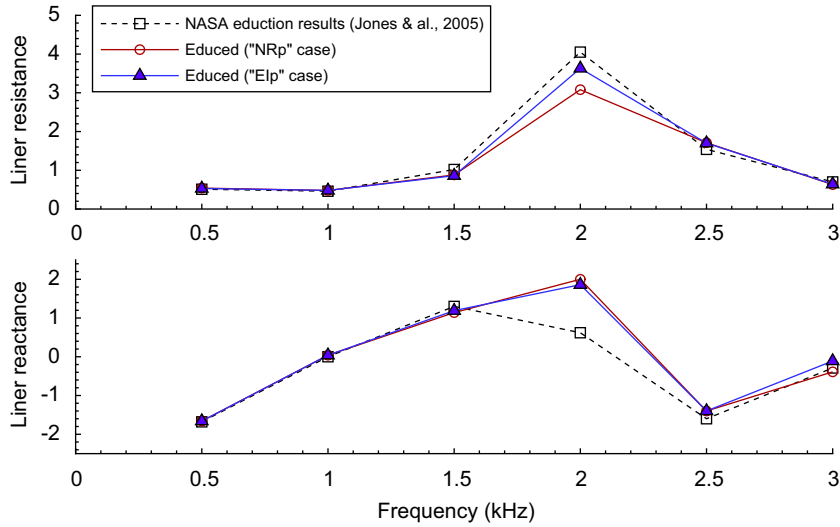


Fig. 9. Liner impedance educed from upper wall pressure measurements on the NASA GIT.

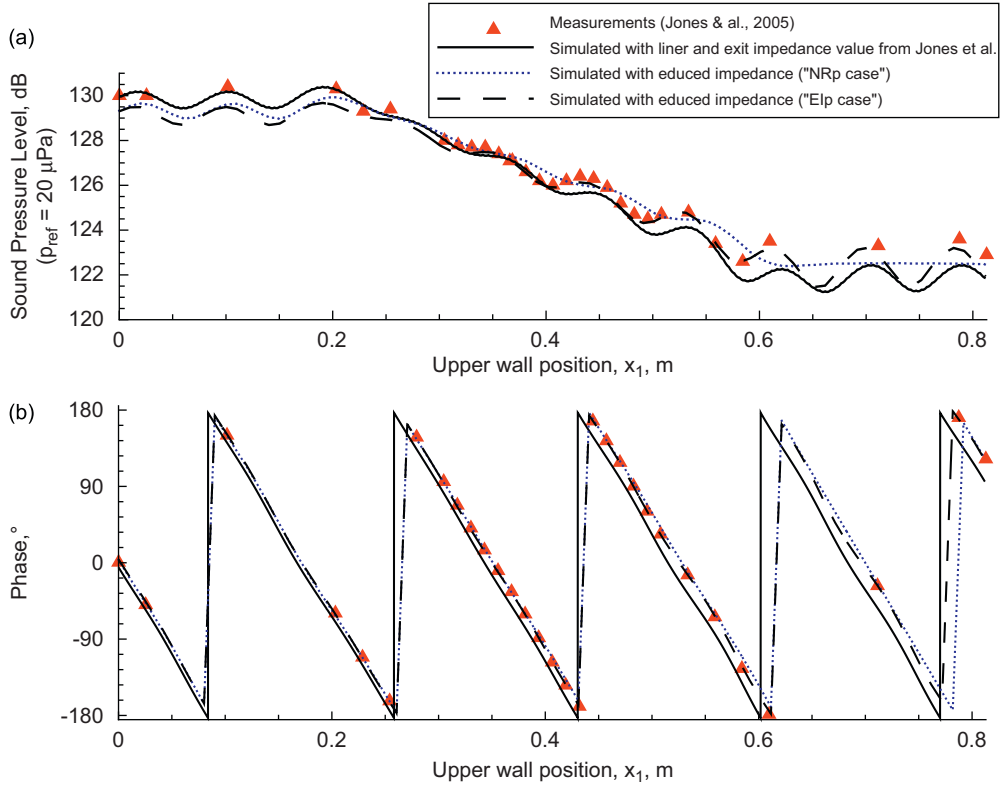


Fig. 10. (a) SPL and (b) phase on the upper wall of the NASA GIT at 2 kHz. Measured values from Ref. [27] (triangular symbols), DG simulation with liner and exit impedance from the reference paper [27] (solid lines), and DG simulation with the impedance educed from the 'NRp' case (dotted lines) and from the 'Elp' case (dashed lines).

Sensitivity of the results to the liner impedance can be investigated by plotting the map of the objective function $\mathcal{J}$ in the impedance plane (r, χ) , normalized by the target value. This quantity will be called "reduced objective function" and denoted by $\mathcal{J}_{\text{red}}$, such as

$$\mathcal{J}_{\text{red}} = \frac{\sum_{m=1}^N \|p_{\text{calc}}^m - p_{\text{meas}}^m\|^2}{\sum_{m=1}^N \|p_{\text{meas}}^m\|^2} \quad (38)$$

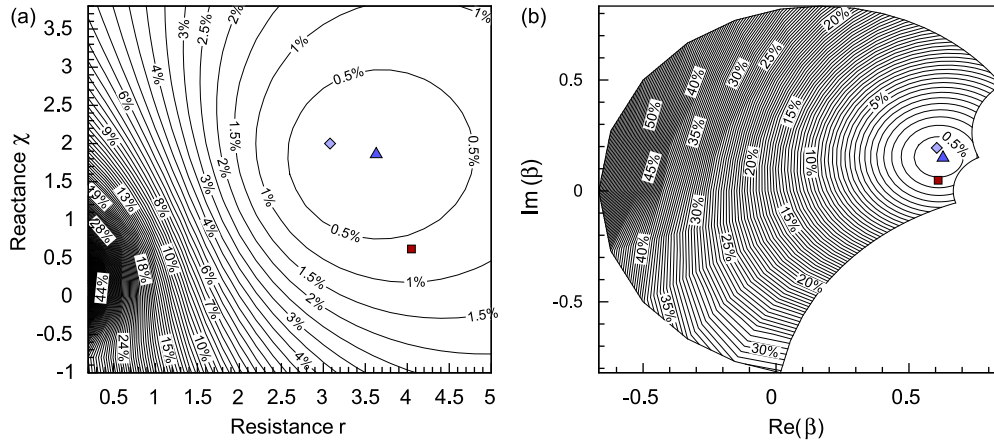


Fig. 11. Objective function based on the acoustic pressure, represented at 2 kHz (a) in the impedance plane (r, χ) and (b) in the reflection coefficient plane ($\text{Re}(\beta), \text{Im}(\beta)$). The exit impedance and calibration coefficient are fixed to their educed values. Symbols locate the educed impedances for the “NRp” case (◆), “Elp” case (▲) and NASA value (■).

The regular mesh chosen here uses steps of $\Delta r = \Delta \chi = 0.2$. The shape of the reduced objective function at 2 kHz (see Fig. 11(a)) shows a relatively shallow region around the minimum, which confirms that a large set of educed impedance values can lead to slightly identical acoustic pressure at the opposite rigid wall. Recall that this plot has been built with our DG code, which places our educed values at the minimum of $\mathcal{J}_{\text{red}}$. Of course, the same figure plot with the NASA FEM code shows the value educed by NASA at the minimum [41]. Our point here is to show the small sensitivity of the objective function to the impedance in this region. For example, for a fixed reactance $\chi = 1.8$, changing the resistance value from $r = 2.6$ to $r = 4.8$ hardly affects the error of $\mathcal{J}$ to the target value, as it remains within the 0.5 percent circle.

This plot has been realized by fixing the exit impedance and the calibration coefficient to their final educed values, which means that only the liner impedance z_l is considered as a parameter. Therefore, the fact that only one, global minimum appears in the objective function must be considered cautiously, as local minima may arise in the complete multiparameter search procedure with z_l , C and z_t .

The same reduced objective function may be displayed in the ($\text{Re}(\beta), \text{Im}(\beta)$) plane (Fig. 11(b)). The same contours levels have been used (0.5 percent between each isocontour ligne), and pictured for the same admissible domain of impedance. It brings to light sharper gradient, which is of particular interest for a fast convergence of the inverse problem. Even though the boundary condition is implemented through a β -formulation (Eq. (24)), the current version of the code is established by considering directly the impedance as a search parameter to ensure positivity of the resistance. However, Fig. 11(b) reveals that it could be particularly advantageous in further developments to treat the reflection coefficient as parameter.

3.2.2. Numerical investigation on eduction from acoustic velocity

The parameter identification method currently being developed is planned to be used with measurement obtained by laser Doppler velocimetry (LDV), and behavior of the eduction process with objective function written in terms of acoustic velocity is worth being explored. The use of adjoint equations makes easy a change of objective function, as it only impacts the second member of the adjoint state equation (31).

As acoustic velocity data on the NASA GIT are not available, a numerical experiment was made. Simulations are performed with the DG code, with input liner and exit impedance values taken from [27]. The synthetic velocity obtained is modified by addition of artificial noise. When dealing with LDV measurements without flow, the main noise contribution lies in the electronic shot noise. To simulate this kind of noise, a normal distribution has been chosen for the streamwise and transversal acoustic velocity, in terms of amplitude and phase. These laws are characterized by their mean values (chosen as the amplitude and phase of the synthetic velocity) and the standard deviation σ_u, σ_v on the amplitudes and ϕ_u, ϕ_v on the phase. These quantities are chosen such as

$$\begin{cases} \sigma_u = \sigma_v = 5 \text{ percent } u_{\text{max}} \\ \phi_u = \phi_v = 5^\circ \end{cases} \quad (39)$$

with u_{max} the maximum amplitude value. At this source level and without flow, the orders of magnitudes taken for the standard deviation on the amplitude and phase are representative of the actual noise in the LDV test-bench of Onera.

The resulting synthesized velocity with added noise is then considered as the “measured” one, $\mathbf{u}_{\text{meas}}$, and is used to calculate the velocity objective function

$$\mathcal{J}(\boldsymbol{\varphi}, z, C) = \int_{\Omega} \|\mathbf{u}_{\text{meas}} - \mathbf{u}_{\text{calc}}\|^2 I_{Q_{\text{obs}}}(\mathbf{x}) \quad (40)$$

This case will be denoted “Elvn” case (for “exit impedance case” from simulated velocity with added noise). Note that contrary to Eq. (36), the squared error is assessed from a vector and integrated on the observation region Ω_{obs} . This one is a 2D area of dimensions $406 \text{ mm} \times 25 \text{ mm}$ just above the liner, as pictured in Fig. 12. It has been chosen to fit the typical zone in which the acoustic velocity is acquired by LDV at Onera.

Figs. 13 and 14 show the educed liner and exit impedances, which compare very well with the input impedances in spite of the added noise. Educed impedances are given in Appendix A (Table A3). They are very close from the impedances

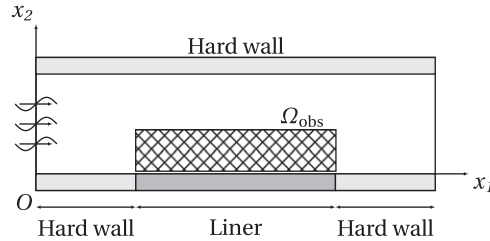


Fig. 12. Observation region for the “Elvn” case.

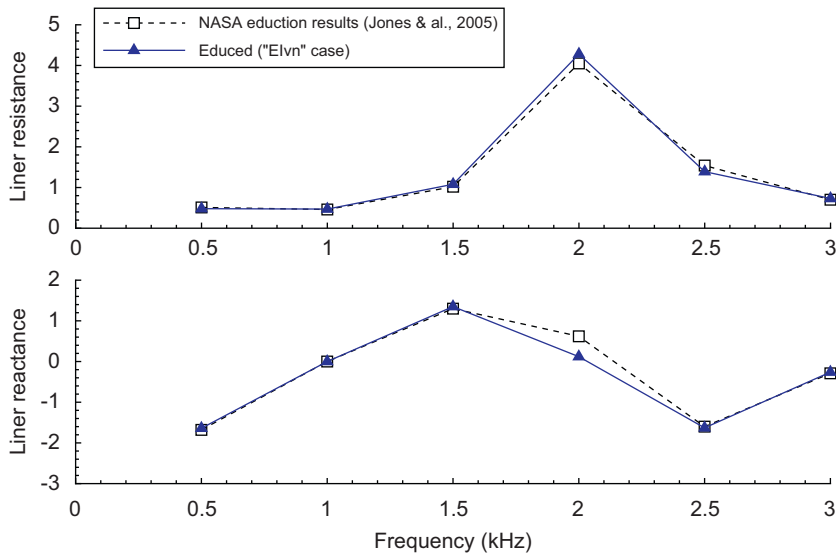


Fig. 13. Liner impedance educed from simulated velocity (“Elvn” case) on the NASA GIT.

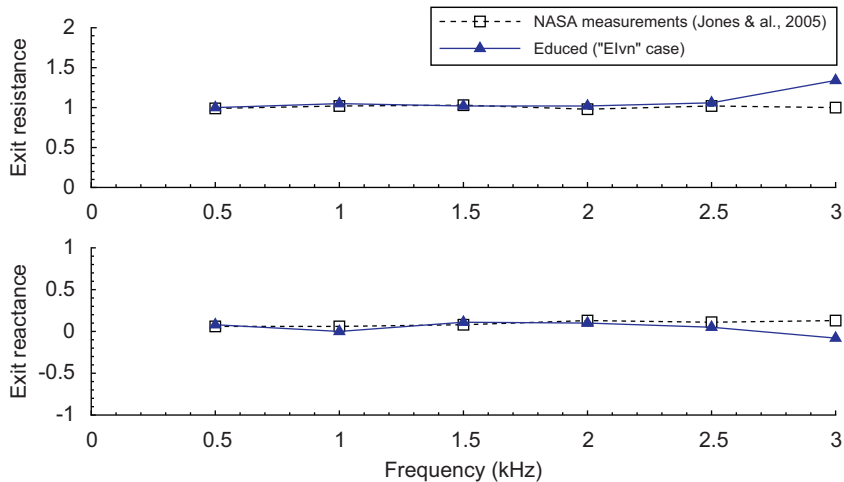


Fig. 14. Exit impedance educed from simulated velocity (“Elvn” case) on the NASA GIT.

of [27], as these latest have been used as input of the code. Only a slight difference is observed on the liner reactance at the anti-resonance frequency (2 kHz), and on the exit resistance and reactance at 3 kHz. This numerical investigation is interesting as it shows that impedance eduction based on 2D acoustic velocity fields above the liner is not only possible, but also show little sensitivity to noise as well as attractive convergence properties as highlighted in the next paragraph.

3.2.3. Influence of the objective function

The eduction method allows effortless evaluation of the impact that can have the choice of objective function. Convergence properties has been compared for pressure and velocity-based objective functions. For the velocity, the “Elvn” case is considered without noise and referenced “Elv”. To remain coherent and avoid any effect of measurement noise, the same approach as in the “Elv” case is applied to pressure. This means that acoustic pressure synthesized by the DG code with impedance values from NASA is used as the target measurement. This case will be denoted “Elp_syn”. The observation region is still the 2D area for “Elv” and the upper wall microphone locations for “Elp_syn”. To allow easy comparison, we chose to represent the reduced objective function as written in (38). In terms of acoustic velocity, this reads

$$\mathcal{J}_{\text{red}} = \frac{\int_{\Omega} \|\mathbf{u}_{\text{meas}} - \mathbf{u}_{\text{calc}}\|^2 I_{\Omega_{\text{obs}}}(\mathbf{x})}{\int_{\Omega} \|\mathbf{u}_{\text{meas}}\|^2 I_{\Omega_{\text{obs}}}(\mathbf{x})} \quad (41)$$

Termination criteria (34) and (35) are ruled by $T_{\mathcal{J}} = 10^{-8}$ and $T_{\nabla} = 10^{-3}$.

The convergence rate of each case is plotted in Fig. 15. Note that the “Elp” case genuinely requires 34 iterations to converge, even if the scale of the graph does not allow to see any change in $\mathcal{J}_{\text{red}}$ after 20 iterations. The search procedure converges more rapidly to the expected value with the velocity-based objective function (40) than with the pressure-based objective function (36). This behavior can be understood as the two components of acoustic velocity are considered on over 250 triangular elements, which represents far more information than the single acoustic pressure on the 31 microphone probes. This clearly appears when the objective function is displayed in the impedance plane (r, χ), as can be seen in Fig. 16 where it has been plotted for the “Elv” and “Elp_syn” cases. Steps of $\Delta r = \Delta \chi = 0.08$ have been taken. Symbols locate the educed impedances which are identical, and therefore superimpose. The velocity-based objective function reveals sharper gradient and a less

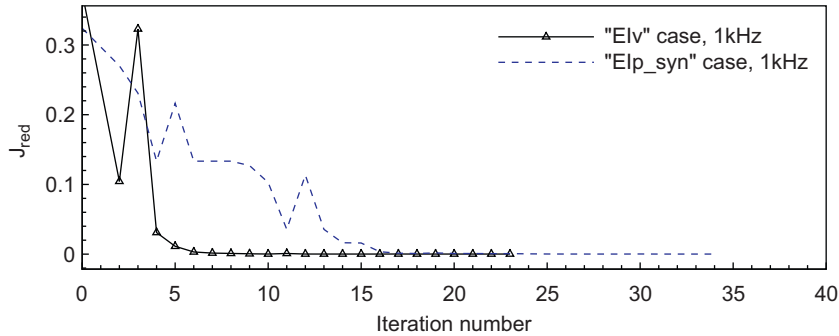


Fig. 15. Evolution of the reduced objective function during the search process at 1 kHz.

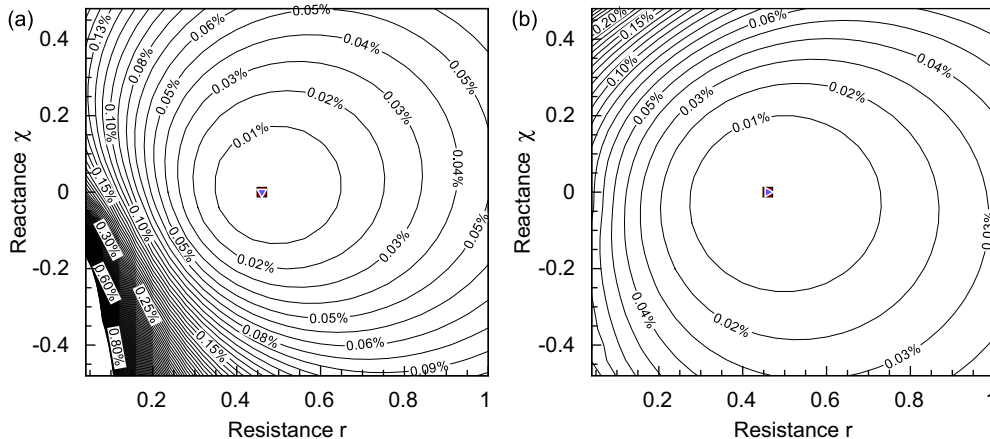


Fig. 16. Objective function at 1 kHz (a) for the “Elv” case and (b) for the “Elp_syn” case. The exit impedance and calibration coefficient are fixed to their educed values. Symbols locate the educed impedances for the “Elv” case (▼), “Elp_syn” case (▴) and NASA value (■).

shallow region around the minimum. This explains the faster convergence in the “Elv” case, but also shows that the uncertainty is less important than with the pressure-based objective function.

3.3. Impedance eduction with liner partitioning

As mentioned earlier (see Section 2.2), a search procedure with several impedances can easily be treated. This allows to go beyond the hypothesis of a liner modeled by a single, homogeneous impedance. It has been done by partitioning the liner in two parts of equal length, with impedances denoted z_{l1} and z_{l2} . The first (complex) search parameter is z_{l1} ; as we expect the two impedances to be quite close from each other, the second search parameter is not directly z_{l2} but the difference $\Delta z_l = z_{l1} - z_{l2}$, which is expected to be small and is thus constrained in $[-0.5; 0.5] + [-1; 1]j$. The configuration is similar to the “NRp” case.

Results on SPL are shown at 0.5 kHz (Fig. 17), where the benefit on the reconstructed pressure is the most visible. No amelioration could be seen on the phase. The educed impedances are $z_{l1} = 0.68 - 1.57j$ and $z_{l2} = 0.38 - 1.88j$, which have to be compared to the impedance educed in the “NRp” case, $z_l = 0.54 - 1.66j$, not far from an average value between z_{l1} and z_{l2} . An increase in the number of parameters thus leads to improvement in the eduction results. Even though further investigation has to be carried out, this result suggests that partitioning can lead to a gain in the accuracy of the educed values.

3.4. Illustration by the adjoint state of the first-order necessary condition for optimality

In most cases of the literature, the adjoint state can directly be interpreted as a sensitivity or receptivity function. Amongst others, Barone and Lele [30] stated that it represents the sensitivity to external forcing of a free mixing-layer flow created downstream of a splitter plate termination. Airiau et al. [29] used it to study the receptivity and sensitivity of an incompressible boundary layer over a flat plate. In the present case, the form of the directional derivative (32) makes difficult an interpretation of the whole adjoint field. Nevertheless, it is interesting to consider the first-order necessary condition for optimality

$$\partial_z \mathcal{J}(\boldsymbol{\varphi}, z, C) = 0 \quad (42)$$

Developing expression (32) in the case of a horizontal liner ($n_1 = 0$ and $n_2 = -1$) leads to

$$-2\beta_l^2 \int_{\Gamma_l} v \bar{v}^* d\Gamma_l = 0 \quad (43)$$

The $\bar{\cdot}$ denotes the complex conjugate. Assuming a liner with a reflection coefficient β_l different from 0 ($z_l \neq 1$), the integral of the product $v \bar{v}^*$ over the liner should be zero. It actually occurs that this is due to the cancelation of the adjoint velocity normal to the liner, i.e. $v^* = 0$ over the lining sample. Once again, this happens only when the first-order necessary

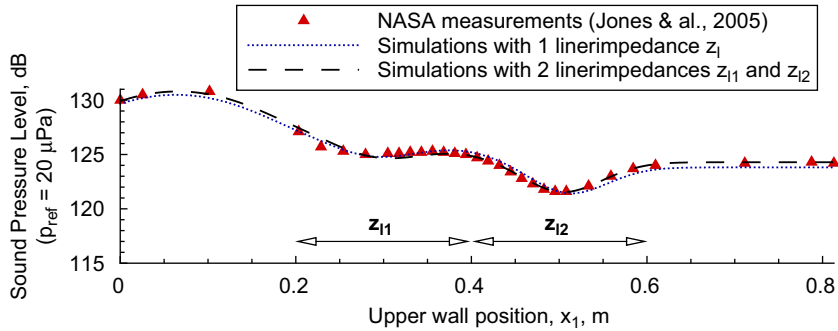


Fig. 17. Reconstructed SPL at 0.5 kHz, with a partitioned liner.

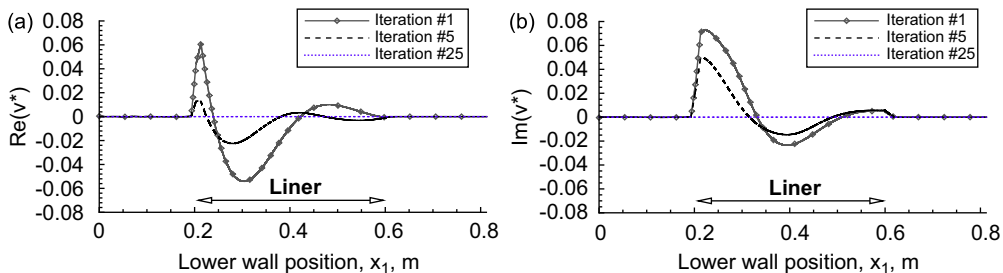


Fig. 18. (a) Real and (b) imaginary part of the transversal adjoint velocity on the lower wall (rigid and lined parts).

condition is satisfied, namely when the eduction process has converged to a suitable value. The evolution of the adjoint velocity at the surface of the liner during the eduction process is shown in Fig. 18, illustrating the fact that v^* tends to zero while the eduction loop progresses. It also allows to check that upstream and downstream of the liner, the rigid wall adjoint condition ($v^* = 0$) is satisfied.

4. Conclusions

A new method for liner impedance eduction has been presented and validated. It seemed important to the authors to describe thoroughly the obtention of the discontinuous Galerkin formulation and adjoint equations, since they differ from the methods usually found in the impedance eduction community. The discontinuous Galerkin scheme is particularly adapted to the adjoint formulation, as the “differentiate-then-discretize” and “discretize-then-differentiate” approaches are equivalent. The main results of validation presented here rely on data published in 2005, directly available in the literature [27]. Simulation results compare very favorably to these measured data, with the possibility to educe exit impedance as well, without need to measure it. Benchmarking with more recent and more complete measurement data [40] is currently being treated.

In particular, it has already been highlighted that the current study is limited to a no-flow configuration, cases in the presence of a grazing flow being planned for publication in a forthcoming paper. A first step has been made in this direction [42], but it still has to be comforted.

Although the results obtained so far concern a configuration without flow, they indicate the powerful resources brought by the process. The analytical expressions of the objective function directional derivatives, obtained through the solution of the adjoint equations, make easy a multiparameter study without additional computational cost. It can be useful for the test of non-homogeneous liners as presented here, or when impedance models such as Biot's, or the extended Helmholtz resonator model in time domain, are employed. Whereas no physical interpretation of the global adjoint field could be found, its behavior during the sequence of process iterations has been exposed. Further calculation may consist in considering the problem as an optimal control problem, in order to get sensitivity of the acoustic field to the liner impedance and thus give information on the best localization of the observation region. The opportunity to displace the observation region towards the lining sample should contribute to get impedances more representative of the actual behavior of the liner. The numerical experiments based on acoustic velocity objective function are promising, and let hope that further benchmarking on LDV measurements will be successful, with a 2D observation region situated above the liner. Another important result is that it could be advantageous, for accuracy as well as for convergence speed, to use the reflection coefficient as a search parameter rather than the impedance, still taking care that the resistance remains positive.

Appendix A. Educed impedances

See (Tables A1–A3).

Table A1

Liner resistance and reactance from the reference paper and educed for the “NRp” case.

Frequency (kHz)	r [27]	χ [27]	r , “NRp” case	χ , “NRp” case
0.5	0.51	–1.68	0.54	–1.66
1.0	0.46	0.00	0.48	0.04
1.5	1.02	1.30	0.88	1.14
2.0	4.05	0.62	3.08	2.00
2.5	1.54	–1.60	1.71	–1.40
3.0	0.70	–0.29	0.63	–0.39

Table A2

Liner resistance and reactance from the reference paper and educed for the “Elp” case.

Frequency (kHz)	r [27]	χ [27]	r , “Elp” case	χ , “Elp” case
0.5	0.51	–1.68	0.53	–1.66
1.0	0.46	0.00	0.48	0.04
1.5	1.02	1.30	0.86	1.19
2.0	4.05	0.62	3.63	1.86
2.5	1.54	–1.60	1.70	–1.40
3.0	0.70	–0.29	0.64	–0.11

Table A3

Liner resistance and reactance from the reference paper and educed for the “Elvn” case.

Frequency (kHz)	r [27]	χ [27]	r , “Elvn” case	χ , “Elvn” case
0.5	0.51	–1.68	0.48	–1.64
1.0	0.46	0.00	0.47	0.00
1.5	1.02	1.30	1.08	1.35
2.0	4.05	0.62	4.28	0.12
2.5	1.54	–1.60	1.39	–1.63
3.0	0.70	–0.29	0.73	–0.26

References

- [1] R.J. Astley, R. Sugimoto, P. Mustafi, Computational aero-acoustics for fan duct propagation and radiation current status and application to turbofan liner optimisation, *Journal of Sound and Vibration* 330 (2011) 3832–3845.
- [2] C. Malmay, S. Carbonne, Y. Aurégan, V. Pagneux, Acoustic impedance measurement with grazing flow, *Proceedings of the Seventh AIAA/CEAS Aeroacoustics Conference*, No. 2001-2193, AIAA, 2001.
- [3] U. Ingard, On the theory and design of acoustic resonators, *Journal of the Acoustical Society of America* 25 (1953) 1037–1061.
- [4] T.H. Melling, The acoustic impedance of perforates at medium and high sound pressure levels, *Journal of Sound and Vibration* 29 (1973) 1–65.
- [5] T. Elnady, H. Bodén, On semi-empirical liner impedance modeling with grazing flow, *Proceedings of the Ninth AIAA/CEAS Aeroacoustics Conference*, No. 2003-3304, AIAA, 2003.
- [6] P.D. Dean, An in situ method of wall acoustic impedance measurement in flow ducts, *Journal of Sound and Vibration* 34 (1974) 97–130.
- [7] M. Lavieille, F. Simon, F. Micheli, Measurement of liner acoustic impedance in a shear layer of a subsonic flow by Laser Doppler Velocimetry, *SAPEM*, 2005.
- [8] W.R. Watson, M.G. Jones, S.E. Tanner, T.L. Parrott, Validation of a numerical method for extracting liner impedance, *AIAA Journal* 34 (1996) 548–554.
- [9] W.R. Watson, M.G. Jones, Validation of a new procedure for impedance eduction in flow, *Proceedings of the 16th AIAA/CEAS Aeroacoustics Conference*, No. 2010-3764, AIAA, 2010.
- [10] W.R. Watson, M.G. Jones, C.H. Gerhold, Implementation and validation of an impedance eduction technique, *Proceedings of the 17th AIAA/CEAS Aeroacoustics Conference*, No. 2011-2867, AIAA, 2011.
- [11] Y. Aurégan, M. Leroux, V. Pagneux, Measurement of liner impedance with flow by an inverse method, *Proceedings of the 10th AIAA/CEAS Aeroacoustics Conference*, No. 2004-2838, AIAA, 2004.
- [12] T. Elnady, H. Bodén, B. Elhadidi, Validation of an inverse semi-analytical technique to educe liner impedance, *AIAA Journal* 47 (2009) 2836–2844.
- [13] C. Richter, Liner Impedance Modeling in the Time Domain with Flow, Ph.D. Thesis, Technischen Universität Berlin, 2009.
- [14] W. Eversman, J.M. Gallman, Impedance eduction with an extended search procedure, *AIAA Journal* 49 (2011) 1960–1970.
- [15] F.Q. Hu, H.L. Atkins, Two-dimensional wave analysis of the discontinuous Galerkin method with non-uniform grids and boundary conditions, *Proceedings of the Eighth AIAA/CEAS Aeroacoustics Conference*, No. 2002-2514, AIAA, 2002.
- [16] H.L. Atkins, Continued development of the discontinuous Galerkin method for computational aeroacoustic applications, *Proceedings of the Third AIAA/CEAS Aeroacoustics Conference*, No. 1997-1581, AIAA, 1997.
- [17] C.R. Vogel, *Computational Methods for Inverse Problems*, SIAM, 2002.
- [18] A. Harten, J.M. Hyman, P.D. Lax, On finite-difference approximations and entropy conditions for shocks, *Communications on Pure and Applied Mathematics* XXIX (1976) 297–322.
- [19] P. Delorme, P. Mazet, C. Peyret, Y. Ventribout, Computational aeroacoustics applications based on a discontinuous Galerkin method, *C.R. Mecanique* 333 (2005) 676–682.
- [20] A. Harten, P.D. Lax, B. van Leer, On upstream differencing and Godunov-type schemes for hyperbolic conservation laws, *SIAM Review* 25 (1983) 35–61.
- [21] J.L. Steger, R.F. Warming, Flux vector splitting of the inviscid gas dynamics equations with application to finite-difference methods, *Journal of Computational Physics* 40 (1981) 263–293.
- [22] G. Gabard, Discontinuous Galerkin methods with plane waves for time-harmonic problems, *Journal of Computational Physics* 225 (2007) 1961–1984.
- [23] E.F. Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics*, second ed. Springer-Verlag, 1999.
- [24] K.W. Thompson, Time dependent boundary conditions for hyperbolic systems, *Journal of Computational Physics* 68 (1987) 1–24.
- [25] M.K. Myers, On the acoustic boundary condition in the presence of flow, *Journal of Sound and Vibration* 71 (1980) 429–434.
- [26] Y. Renou, Y. Aurégan, Failure of the Ingard–Myers boundary condition for a lined duct: an experimental investigation, *Journal of the Acoustical Society of America* 130 (2011) 52–60.
- [27] M.G. Jones, W.R. Watson, T.L. Parrott, Benchmark data for evaluation of aeroacoustic propagation codes with grazing flow, *Proceedings of the 11th AIAA/CEAS Aeroacoustics Conference*, No. 2005-2853, AIAA, 2005.
- [28] L. Wang, D.J. Mavriplis, W.K. Anderson, Adjoint sensitivity formulation for discontinuous Galerkin discretizations in unsteady inviscid flow problems, *AIAA Journal* 48 (2010) 2867–2883.
- [29] C. Airiau, S. Walther, A. Bottaro, Boundary layer sensitivity and receptivity, *C.R. Mecanique* 330 (2002) 259–265.
- [30] M.F. Barone, S.K. Lele, Receptivity of the compressible mixing layer, *Journal of Fluid Mechanics* 540 (2005) 301–335.
- [31] J.B. Freund, Adjoint-based optimization for understanding and suppressing jet noise, *Journal of Sound and Vibration* 330 (2011) 4114–4122.
- [32] M. Wei, J.B. Freund, A noise-controlled free shear flow, *Journal of Fluid Mechanics* 546 (2006) 123–152.
- [33] M.D. Gunzburger, *Perspectives in Flow Control and Optimization*, SIAM, 2003.
- [34] Y. Ventribout, Contrôle des perturbations aéroacoustiques par impédances de parois: application à un modèle de matériaux poreux [Wall Impedance Control of Aeroacoustic Perturbations: Application to a Porous Material], Ph.D. Thesis, Ecole Nationale Supérieure de l’Aéronautique et de l’Espace, 2006.
- [35] C. Zhu, R.H. Byrd, P. Lu, J. Nocedal, L-BFGS-B: Algorithm 778: L-BFGS-B, FORTRAN routines for large scale bound constrained optimization, *ACM Transactions on Mathematical Software* 23 (1997) 550–560.
- [36] R.H. Byrd, P. Lu, J. Nocedal, A limited memory algorithm for bound constrained optimization, *SIAM Journal on Scientific and Statistical Computing* 16 (1995) 1190–1208.
- [37] J. Primus, F. Simon, E. Piot, Validation of a direct propagation model for liner impedance eduction, *Proceedings of the 17th AIAA/CEAS Aeroacoustics Conference*, No. 2011-2869, AIAA, 2011.
- [38] Y. Özyörük, L.N. Long, M.G. Jones, Time-domain numerical simulation of a flow-impedance tube, *Journal of Computational Physics* 146 (1998) 29–57.
- [39] W.R. Watson, M.G. Jones, T.L. Parrott, Investigation of an anomaly observed in impedance eduction techniques, *Proceedings of the 14th AIAA/CEAS Aeroacoustics Conference*, No. 2008-3013, AIAA, 2008.

- [40] M.G. Jones, W.R. Watson, On the use of experimental methods to improve confidence in reduced impedance, *Proceedings of the 17th AIAA/CEAS Aeroacoustics Conference*, No. 2011-2865, AIAA, 2011.
- [41] W.R. Watson, M.G. Jones, Explanation of anomalous behavior observed in impedance reduction techniques using measured data, *Proceedings of the 16th AIAA/CEAS Aeroacoustics Conference*, No. 2010-3766, AIAA, 2010.
- [42] E. Piot, J. Primus, F. Simon, Liner impedance reduction technique based on velocity fields, *Proceedings of the 18th AIAA/CEAS Aeroacoustics Conference*, No. 2012-2198, AIAA, 2012.